

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 16

MULTIPLICATION OF TWO 16 – BIT NUMBERS

AIM: To write and execute an 8085 assembly language program to multiply two 16 – bit numbers and store the result in memory.

ALGORITHM

1. Load the first 16-bit number into AX.
2. Load the second 16-bit number into BX.
3. Multiply AX by BX using MUL BX.
4. The 32-bit result is stored in DX:AX.
5. Store the result in memory.
6. Stop the program.

CODE:

```
MOV AX,0012H
```

```
MOV BX,0004H
```

```
MUL BX
```

```
MOV [8050H],AX
```

```
MOV [8052H],DX
```

```
HLT
```

OUTPUT:

